BD-808 Bass Drum Module

Rev. 1

Description

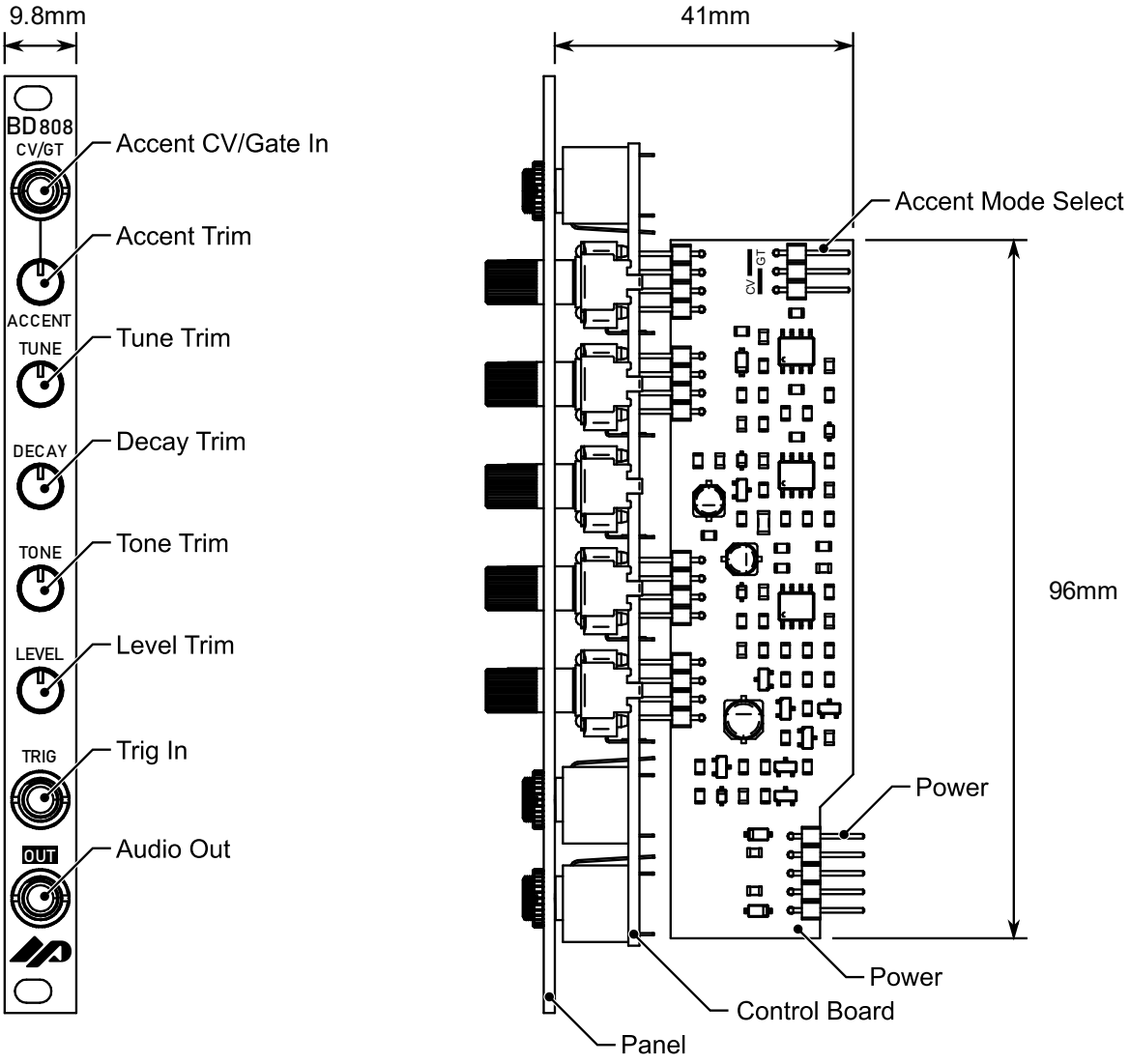
The BD-808 is a clone of the TR-808 Bass Drum, modified to fit the Eurorack size format and voltage requirements. It supports additional functionality not available in the original TR-808 drum machine, including features that make it more suited to the modular synthesizer format.

The BD-808 module fits in a compact 2hp form factor to minimize used rack space and has a max depth of 41 mm with recessed power header to fit almost any rack size. All inputs are robust with overvoltage and reverse-voltage protection.

Features

- Compact 2hp form factor
- Internal gate to timed trigger conversion
- Dual selectable Accent (AC) input modes
    - AC Control Voltage (CV) mode
    - AC Gate (GT) mode
- Accent level trim
- Tune, Decay, Tone trim
- Input overvoltage and reverse-polarity protection
- Power input reverse-polarity protection
- Current Draw: +12V: 15mA, -12V: 12mA

Module Layout



Functionality

Trig

The Trig input takes gate/trig inputs longer than 1ms in duration and converts them into 1ms timed pulses. This avoids any need for an external gate to trigger converter and prevents double-triggering on the falling edge of longer gate inputs. Note that input pulses shorter than 1ms are not altered. Trig inputs can be any voltage between 5V and 12V.

Tune

The Tune trim sets the pitch of the drum sound. The pitch can range from a high 'tom' to a deep bass sound.

Decay

The Decay trim allows the decay length of the drum sound to be lengthened or shortened.

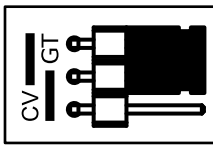
Tone

The Tone trim adjusts the output filter of the drum to produce a snappier kick sound or a smoother bass sound.

Accent

The Accent (AC) effect adds a dynamic feel to a drum sequence by modifying the strength of the drum hit (i.e. volume/tone). The Accent effect can be set by the Accent trim alone (with no physical inputs to the AC CV/GT jack), or with external voltages. The BD-808 has two accent modes for external voltages, selectable with a jumper and 3-pin mode-select header: AC CV Mode and AC GT Mode.

AC CV Mode

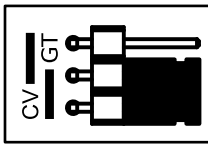


AC CV Mode Jumper Position

AC CV Mode allows the Accent setting to be modified by an external analog CV input. This input can range from 0V to 6V, with 0V being a 'light' drum hit and 6V being a 'hard' hit. CV input voltages larger than 6V will not damage the module, however they will also not further increase the accent setting.

In AC CV mode the Accent trim acts as an offset for the CV input. The Accent trim setting will always be active and applied to every beat, regardless of the presence of a CV voltage.

AC GT Mode



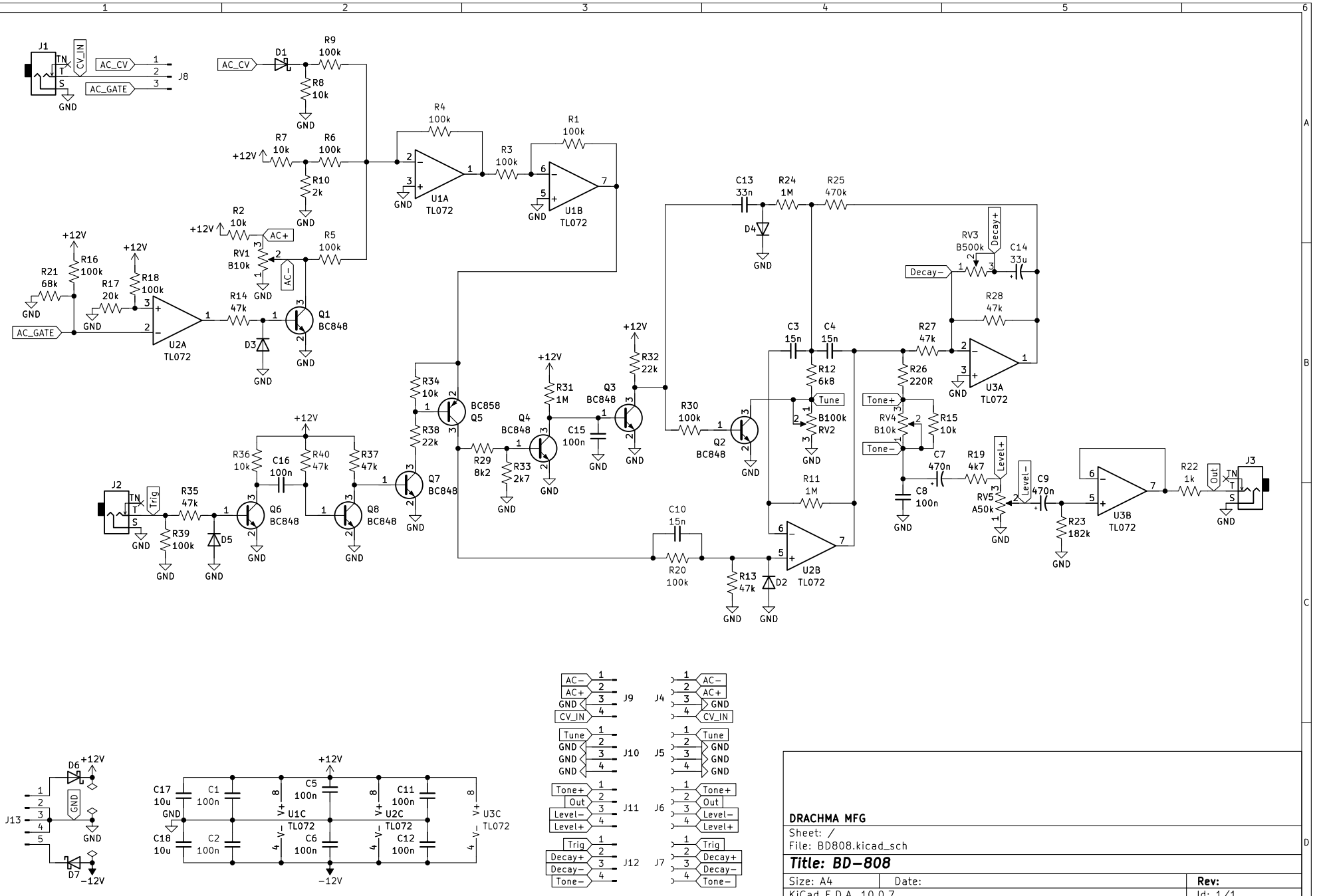
AC GT Mode Jumper Position

AC GT mode allows the Accent effect to be gated (On/Off) rather than continuously variable, similar to the functionality of the original TR-808. To gate the Accent effect (i.e., turn it On), any voltage input between 5V and 12V (High) can be used. A voltage input less than 2V (Low) will turn the Accent effect Off.

Though the AC GT input should be treated like any other digital gate input, note that it is internally pulled up to +5V. Floating input signals will be treated as a High input, allowing for the use of the Accent trim with no physical inputs to the AC CV/GT jack. In order to leave beats un-accented in AC GT Mode, the AC GT input must be pulled down to G

Bill of Materials

Value	Qty.	Name on Board	Description	Footprint
220Ω	1	R26	220Ω 1% 0.125W Resistor	0805
1kΩ	1	R22	1kΩ 1% 0.125W Resistor	0805
2kΩ	1	R10	2kΩ 1% 0.125W Resistor	0805
2.7kΩ	1	R33	2.7kΩ 1% 0.125W Resistor	0805
4.7kΩ	1	R19	4.7kΩ 1% 0.125W Resistor	0805
6.8kΩ	1	R12	6.8kΩ 1% 0.125W Resistor	0805
8.2kΩ	1	R29	8.2kΩ 1% 0.125W Resistor	0805
10kΩ	6	R2, R7, R8, R15, R34, R36	10kΩ 1% 0.125W Resistor	0805
20kΩ	1	R17	20kΩ 1% 0.125W Resistor	0805
22kΩ	2	R32, R38	22kΩ 1% 0.125W Resistor	0805
47kΩ	7	R13, R14, R27, R28, R35, R37, R40	47kΩ 1% 0.125W Resistor	0805
68kΩ	1	R21	68kΩ 1% 0.125W Resistor	0805
100kΩ	11	R1, R3, R4, R5, R6, R9, R16, R18, R20, R30, R39	100kΩ 1% 0.125W Resistor	0805
182kΩ	1	R23	182kΩ 1% 0.125W Resistor	0805
470kΩ	1	R25	470kΩ 1% 0.125W Resistor	0805
1MΩ	3	R11, R24, R31	1MΩ 1% 0.125W Resistor	0805
15nF	3	C3, C4, C10	15nF 50V C0G MLCC	0805
33nF	1	C13	33nF 50V C0G MLCC	0805
100nF	1	C8	100nF 50V C0G MLCC	1206
100nF	8	C1, C2, C5, C6, C11, C12, C15, C16	100nF 50V X7R MLCC	0805
470nF	2	C7, C9	470nF 50V Elec. Cap.	4x5.4mm SMD
10uF	2	C17, C18	10uF 50V X5R MLCC	0805
33uF	1	C14	33uF 25V Elec. Cap.	5x5.7mm SMD
1N4148 Diode	4	D2, D3, D4, D5	1N4148 Gen. Purp. Diode	SOD-323
Schottky Diode	3	D1, D6, D7	Schottky Diode 40V	SOD-123
BC846 NPN	7	Q1, Q2, Q3, Q4, Q6, Q7, Q8	BC846 NPN Transistor	SOT-23
BC856 PNP	1	Q5	BC856 PNP Transistor	SOT-23
TL072	3	U1, U2, U3	TL072 Dual Op Amp	SOIC-8 150mil
B10k Pot Lin	2	RV1, RV4	10k Lin. Potentiometer	Alpha 9mm vertical
B100k Pot Lin	1	RV2	100k Lin. Potentiometer	Alpha 9mm vertical
B500k Pot Lin	1	RV3	500k Lin. Potentiometer	Alpha 9mm vertical
A50K Pot Log	1	RV5	50k Log. Potentiometer	Alpha 9mm vertical
3.5mm Jack	3	J1, J2, J3	3.5mm Mono Jack	PJ-3001F
5-Pin Header	1	J13	2.54mm Header 5x1	2.54mm 5-pin right angle
4-Pin Header	4	J4/J9, J5/J10, J6/J11, J7/J12	2.54mm Header 4x1	2.54mm 4-pin right angle
3-Pin Header	1	J8	2.54mm Header 3x1	2.54mm 3-pin right angle
2-Pos. Jumper	1	N/A (For use on J8)	2.54mm 2-Pos. Jumper	N/A


J1
CV_IN
AC_CV
AC_GATE
J8
GND
D1
R9
100k
R8
10k
R4
100k
R7
10k
R6
100k
+12V
R10
2k
R3
100k
R1
100k
U1A
TL072
U1B
TL072
R2
10k
AC+
RV1
B10k
AC−
R5
100k
R21
68k
R16
100k
R18
100k
R17
20k
R14
47k
Q1
BC848
D3
U2A
TL072
C13
33n
R24
1M
R25
470k
D4
RV3
B500k
Decay+
Decay−
C14
33u
R28
47k
C3
15n
C4
15n
R27
47k
R26
220R
R12
6k8
Tune
B100k
RV2
U3A
TL072
Tone+
RV4
B10k
Tone−
R15
10k
C7
470n
R19
4k7
Level+
Level−
C9
470n
C8
100n
RV5
A50k
R23
182k
U3B
TL072
R22
1k
Out
J3
R32
22k
R34
10k
BC858
Q5
R31
1M
Q3
BC848
R38
22k
Q4
BC848
C15
100n
R30
100k
Q2
BC848
R29
8k2
R33
2k7
R36
10k
C16
100n
R40
47k
R37
47k
Q7
BC848
J2
Trig
R35
47k
R39
100k
D5
Q6
BC848
Q8
BC848
C10
15n
R20
100k
R11
1M
R13
47k
D2
U2B
TL072
D6
D7
J13
C17
10u
C18
10u
C1
100n
C2
100n
C5
100n
C6
100n
C11
100n
C12
100n
U1C
TL072
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J10
J11
J12
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J5
J6
J7
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